

Control Systems A

Week 3 - Transfer Functions and the s -Plane

UoN - FEE 3411

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What we are doing today

- **Hour 1 — getting a transfer function.** The Laplace transform as a way of solving differential equations, the definition of the transfer function, and the impedance shortcut promised at the end of Week 2.
- **Hour 2 — reading one.** Poles, zeros, the pole–zero map, d.c. gain, and the two extensions the syllabus asks for: discrete-time systems and pure time delay.

Why this matters

Week 2 built models. This week turns them into the object every remaining week of FEE3411 actually operates on — and draws the picture, the s -plane, that Weeks 5–12 are all read off.

Syllabus: Laplace transform solution to a linear constant-coefficient differential equation. Transfer functions as developed from system differential equations. Definitions and properties in the s -plane. . . Continuous and discrete time-invariant systems. Delay time. ELO 2, ELO 4.

Why the Laplace transform

Solving differential equations the hard way

Time-domain solution of $a_n y^{(n)} + \dots + a_0 y = b_m u^{(m)} + \dots + b_0 u$ means: complementary function, then particular integral, then fit the constants — *a fresh piece of work for every new input*.

The Laplace transform replaces all of it with algebra:

$$F(s) = \mathcal{L}\{f(t)\} = \int_{0^-}^{\infty} f(t) e^{-st} dt, \quad s = \sigma + j\omega.$$

The lower limit is 0^- — just *before* $t = 0$ — so an impulse at the origin is captured whole.

Khalil, App. A, p. 438.

The properties that do the work

Pairs

$\delta(t)$	1
$1(t)$	$1/s$
t	$1/s^2$
e^{-at}	$1/(s+a)$
$\sin \omega t$	$\omega/(s^2 + \omega^2)$
$\cos \omega t$	$s/(s^2 + \omega^2)$
$e^{-at} \sin \omega t$	$\frac{\omega}{(s+a)^2 + \omega^2}$
$e^{-at} \cos \omega t$	$\frac{s+a}{(s+a)^2 + \omega^2}$

Fuller table: Khalil, App. C, p. 444.

Properties

- $\dot{f} \rightarrow sF(s) - f(0^-)$
- $\ddot{f} \rightarrow s^2 F(s) - sf(0^-) - \dot{f}(0^-)$
- $\int_0^t f \rightarrow F(s)/s$
- $f(t-\tau)1(t-\tau) \rightarrow e^{-\tau s} F(s)$
- $f_1 * f_2 \rightarrow F_1 F_2$
- $f(\infty) = \lim_{s \rightarrow 0} sF(s)$ — if stable

What the transform buys

Key idea

Three things at once: differentiation becomes multiplication by s ; convolution becomes multiplication; and the initial conditions are carried into the algebra automatically instead of being fitted at the end.

The price is that the answer comes back in the s -domain and has to be inverted — which is what partial fractions are for.

Every derivative becomes a power of s ; every initial condition becomes an additive constant. A differential equation in t becomes a *polynomial* equation in s .

Solving an ODE by Laplace

The four steps

- 1 Transform every term, keeping the initial conditions.
- 2 Collect in $Y(s)$ and solve — ordinary algebra.
- 3 Split $Y(s)$ into partial fractions.
- 4 Invert term by term from the table.

Common pitfall

Step 1 is where marks are lost. $\mathcal{L}\{\ddot{y}\} = s^2 Y - s y(0^-) - \dot{y}(0^-)$ — both initial conditions, with the right powers of s . Dropping them silently turns the problem into a different one.

Worked example: an ODE with non-zero initial conditions

Solve $\ddot{y} + 3\dot{y} + 2y = 2u(t)$, $u = 1(t)$, with $y(0^-) = 2$, $\dot{y}(0^-) = 0$.

Transform and collect:

$$[s^2 Y - 2s] + 3[sY - 2] + 2Y = \frac{2}{s} \quad \Rightarrow \quad Y(s)(s^2 + 3s + 2) = \frac{2}{s} + 2s + 6.$$

Split the answer in two

$$Y(s) = \underbrace{\frac{2}{s(s+1)(s+2)}}_{\text{forced}} + \underbrace{\frac{2s+6}{(s+1)(s+2)}}_{\text{natural}}$$

$s^2 + 3s + 2 = (s+1)(s+2)$ is the *characteristic polynomial* of Week 2 — it is the denominator of *both* terms, whatever the input and whatever the initial conditions.

Invert, and read the answer

Partial fractions give

$$Y(s) = \frac{1}{s} + \frac{2}{s+1} - \frac{1}{s+2} \quad \Rightarrow \quad \boxed{y(t) = 1 + 2e^{-t} - e^{-2t}}, \quad t \geq 0.$$

Check: $y(0) = 2 \checkmark$ $\dot{y}(0) = 0 \checkmark$ $y(\infty) = 1 \checkmark$

Key idea

The *shape* of the natural response — which exponentials appear — is set by the characteristic equation, i.e. by the plant. The *size* of each term is set by the input and the initial conditions. That separation is the whole reason the s-plane picture works.

The transfer function

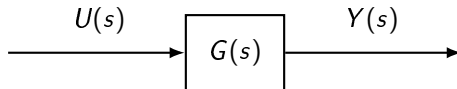
Definition

Initial conditions describe how the system happened to be *started*, not what the system *is*. So set them all to zero.

Key idea

$$G(s) = \left. \frac{Y(s)}{U(s)} \right|_{\text{zero initial conditions}}$$

The transfer function is a property of the system alone: it does not depend on the input, and it does not depend on how the system was started.



Cf. Nise §2.3, p. 44, and Fig. 2.2, p. 45.

From the differential equation, in one line

Applying the definition to Week 2's general model:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b_ms^m + b_{m-1}s^{m-1} + \dots + b_1s + b_0}{a_ns^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0} = \frac{N(s)}{D(s)}.$$

Reading it in reverse is just as useful: given $G(s)$, write the differential equation straight back down by replacing $s^k Y$ with $y^{(k)}$ and $s^k U$ with $u^{(k)}$.

Common pitfall

The zero-initial-condition assumption is part of the definition, not an approximation — but it does not make initial conditions unimportant. A transfer function says how the system responds to an *input*, never how it decays from a *state*. Non-zero initial conditions send you back to the previous section's method.

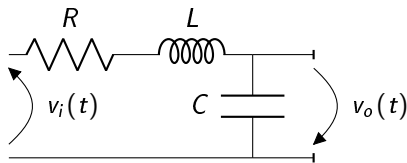
Five properties, all used later

- **The denominator is the characteristic polynomial.** $D(s) = 0$ is Week 2's characteristic equation. Every stability method from Week 7 on is a statement about its roots.
- **Order** $= n$, the degree of $D(s)$.
- **Proper / strictly proper.** Physical systems have $m \leq n$; most have $m < n$. The relative degree $n - m$ shows up in the very first instant of the step response.
- **The impulse response.** $\mathcal{L}\{\delta\} = 1$, so $Y(s) = G(s)$: the transfer function *is* the impulse response, transformed.
- **Cascade.** Blocks in series multiply, $Y = G_2 G_1 U$ — the property Week 4 builds all of block-diagram algebra on.

The impedance shortcut

Worked example: the series RLC network

Week 2 got the differential equation $LC\ddot{v}_o + RC\dot{v}_o + v_o = v_i$ from KVL. Transform it with zero initial conditions:



$$\frac{V_o(s)}{V_i(s)} = \frac{1}{LCs^2 + RCs + 1}.$$

With $R = 3\ \Omega$, $L = 1\ \text{H}$, $C = 0.5\ \text{F}$: $G(s) = 2/[(s+1)(s+2)]$. Roots -1 and -2 — distinct, real, negative, so two decaying exponentials and no oscillation. D.c. gain $G(0) = 1$.

Impedances: skip the differential equation entirely

Transforming Week 2's element relations with zero initial conditions gives $V = ZI$ for every element — Ohm's law with a complex impedance:

$$Z_R = R, \quad Z_L = Ls, \quad Z_C = \frac{1}{Cs}.$$

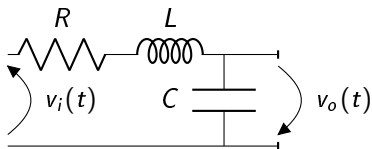
Key idea

Impedances add in series and combine reciprocally in parallel, exactly as resistances do. So an s-domain network is reduced by ordinary resistive circuit analysis — voltage dividers, current dividers, series/parallel — and the transfer function drops out with no differential equation ever written.

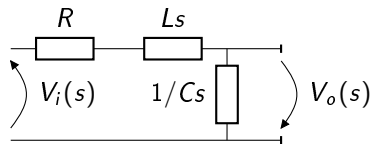
For the RLC network the output is across Z_C in a series loop, so one voltage divider finishes it:

$$\frac{V_o}{V_i} = \frac{Z_C}{Z_R + Z_L + Z_C} = \frac{1/Cs}{R + Ls + 1/Cs} = \frac{1}{LCs^2 + RCs + 1}.$$

The network, drawn both ways



(a) time domain



(b) s-domain impedances

Common pitfall

The shortcut assumes *zero initial conditions* — no initial inductor current, no initial capacitor charge. That is exactly the transfer-function condition, so it is always available for transfer-function work. Given an initial capacitor voltage and asked for the full response, go back to the long method.

Poles, zeros and the pole-zero map

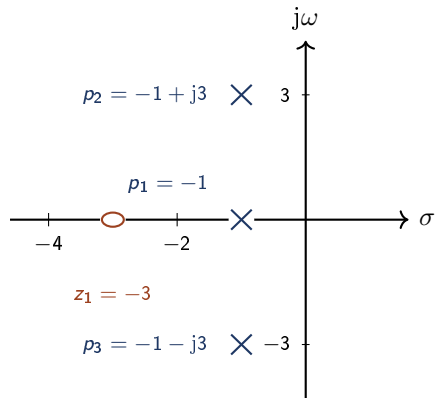
Factored form

$$G(s) = K \frac{(s - z_1)(s - z_2) \cdots (s - z_m)}{(s - p_1)(s - p_2) \cdots (s - p_n)}$$

- **Zeros** z_i — roots of the numerator; $G(z_i) = 0$.
- **Poles** p_i — roots of the denominator; $G(s) \rightarrow \infty$. These are the roots of the characteristic equation.
- K is the **gain factor**. It is *not* the d.c. gain.

Real coefficients $\Rightarrow$ complex poles and zeros come in conjugate pairs $\Rightarrow$ **the pole-zero map is always symmetric about the real axis**. Poles are drawn $\times$, zeros $\circ$.

A pole-zero map



$$G(s) = 4(s + 3)/[(s + 1)(s^2 + 2s + 10)]. \text{ Cf. Nise §4.2, p. 162.}$$

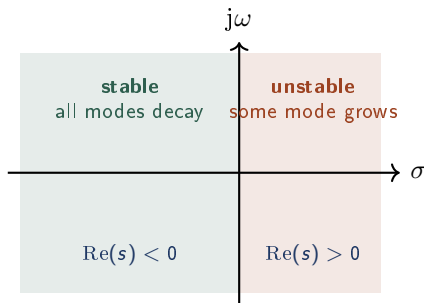
The poles set the natural modes

One pole, one mode

Partial fractions give one term per pole; each inverts to a characteristic function of time. **Real part sets growth or decay; imaginary part sets oscillation.**

Pole location	Term in $y(t)$	Shape
Negative real, $s = -a$	Ae^{-at} , decaying	
At the origin, $s = 0$	A , constant (an integrator)	
Positive real, $s = +a$	Ae^{+at} , growing	
Complex pair, $s = -a \pm j\omega$	$Ae^{-at} \sin(\omega t + \phi)$, damped	
Pair on the $j\omega$ axis	$A \sin(\omega t + \phi)$, sustained	
Complex pair, $s = +a \pm j\omega$	$Ae^{+at} \sin(\omega t + \phi)$, growing	

The stability picture



Key idea

A mode decays if and only if its pole has a *strictly negative real part*. So an LTI system is stable exactly when **all** its poles lie in the open left half plane.

A pole *on* the $j\omega$ axis is marginally stable: it neither decays nor grows.

Every stability test in Weeks 7–12 — Routh–Hurwitz, root locus, Nyquist — is a different way of answering one question: *are any poles to the right of the imaginary axis?*

Inverse Laplace by partial fractions

Three cases cover everything in this unit

- **Distinct real poles.** One term per pole; residues by the cover-up rule — multiply by the factor, evaluate at the pole.
- **Repeated poles.** $(s + a)^r$ needs r terms, $A_1/(s + a) + \dots + A_r/(s + a)^r$, inverting to e^{-at} , te^{-at} , ..., $t^{r-1}e^{-at}$.
- **Complex-conjugate poles.** Keep the quadratic $s^2 + 2\zeta\omega_n s + \omega_n^2$ whole and *complete the square*.

Common pitfall

Do not split a complex pair into two first-order terms with complex residues unless you mean to recombine them. It is legitimate but error-prone, and leaves you with complex exponentials still to turn back into real sines and cosines. Completing the square keeps everything real throughout.

Worked example: a complex-pole step response

$G(s) = \frac{10}{s^2 + 2s + 10}$. Poles: $s = \frac{-2 \pm \sqrt{4 - 40}}{2} = -1 \pm j3$ — a left-half-plane pair, so a damped oscillation. Matching $s^2 + 2\zeta\omega_n s + \omega_n^2$: $\omega_n = \sqrt{10} = 3.16 \text{ rad s}^{-1}$, $\zeta = 1/\sqrt{10} = 0.316$.

Expand, then complete the square

$$Y(s) = \frac{10}{s(s^2 + 2s + 10)} = \frac{1}{s} - \frac{s + 2}{(s + 1)^2 + 3^2},$$
$$\frac{s + 2}{(s + 1)^2 + 3^2} = \frac{s + 1}{(s + 1)^2 + 3^2} + \frac{1}{3} \cdot \frac{3}{(s + 1)^2 + 3^2}$$

$$y(t) = 1 - e^{-t} \left(\cos 3t + \frac{1}{3} \sin 3t \right)$$

Check: $y(0) = 0$ ✓ $y(\infty) = 1 = G(0)$ ✓ Envelope decays as e^{-t} (real part); ringing at 3 rad s^{-1} (imaginary part).

What zeros do

One zero, three systems

All three have poles at $-1, -2$ and d.c. gain 1. Only the zero differs:

$$G_1 = \frac{2}{(s+1)(s+2)}, \quad G_2 = \frac{2}{3} \frac{s+3}{(s+1)(s+2)}, \quad G_3 = \frac{2}{3} \frac{3-s}{(s+1)(s+2)}.$$

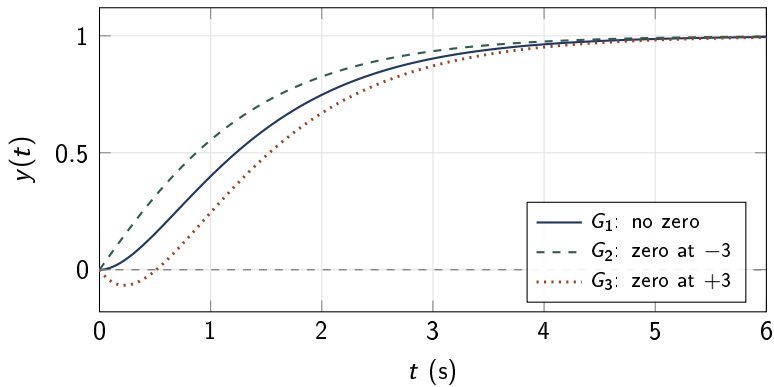
$$y_1 = 1 - 2e^{-t} + e^{-2t}, \quad y_2 = 1 - \frac{4}{3}e^{-t} + \frac{1}{3}e^{-2t}, \quad y_3 = 1 - \frac{8}{3}e^{-t} + \frac{5}{3}e^{-2t}.$$

The same two exponentials in all three — only the coefficients differ. But the starts are completely different:

$$\dot{y}_1(0) = 0, \quad \dot{y}_2(0) = +\frac{2}{3}, \quad \dot{y}_3(0) = -\frac{2}{3}.$$

G_3 leaves in the *wrong direction*, dipping to $-1/15$ at $t = \ln(5/4) = 0.223$ s before settling at 1.

Seen on a step response



Zeros reweight; they do not create

Key idea

A zero adds no new mode and changes no time constant — the exponentials are identical in all three responses. What it changes is the *weighting* of the modes, and so the shape of the transient. A right-half-plane zero gives an initial response in the wrong direction: a **non-minimum-phase** system.

Common pitfall

A right-half-plane *zero* does not cause instability — G_3 settles perfectly well at 1. Only right-half-plane *poles* do. What the zero does is make the system hard to control, because feedback reacting to the initial wrong-way motion pushes the wrong way. It returns in Week 8 as the characteristic shape of a non-minimum-phase root locus.

D.c. gain and the final value theorem

The steady-state gain

For a constant input all derivatives vanish, so $s \rightarrow 0$:

$$G(0) = \frac{b_0}{a_0},$$

the ratio of the constant terms. For a unit step, the steady-state output is $G(0)$. Generally, $y(\infty) = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} sG(s)U(s)$.

Common pitfall

The final value theorem is valid only if the final value *exists* — all poles of $sY(s)$ strictly in the left half plane. Applied blindly to $G(s) = 1/(s - 1)$ it returns $y(\infty) = G(0) = -1$, while the true step response $y(t) = e^t - 1$ grows without bound. Check stability first: the theorem cannot detect its own misuse.

And $K \neq G(0)$: for G_2 above, $K = 2/3$ but $G_2(0) = \frac{2}{3} \cdot \frac{3}{2} = 1$.

Discrete systems and time delay

Discrete time-invariant systems

A digital controller sees samples $u[k] = u(kT)$, and its model is a **difference** equation:

$$y[k] + \alpha_1 y[k-1] + \dots = \beta_0 u[k] + \dots$$

The z -transform plays the Laplace transform's role: with zero initial conditions a one-step delay $y[k-1]$ becomes $z^{-1}Y(z)$, giving a **pulse transfer function** $G(z) = Y(z)/U(z)$.

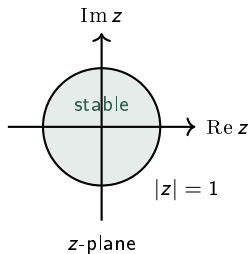
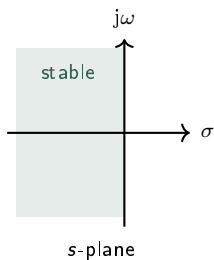
Worked example

$$y[k] = 0.5 y[k-1] + u[k] \Rightarrow Y(z)(1 - 0.5z^{-1}) = U(z), \text{ so}$$

$$G(z) = \frac{1}{1 - 0.5z^{-1}} = \frac{z}{z - 0.5},$$

one pole at $z = 0.5$. Step response $y[k] = 2 - (0.5)^k$: 1, 1.5, 1.75, 1.875, $\dots \rightarrow G(1) = 2$, the d.c. gain evaluated at $z = 1$ rather than $s = 0$.

Two planes, one idea



Key idea

Continuous: stable when every pole has $\text{Re}(s) < 0$.

Discrete: stable when every pole has $|z| < 1$.

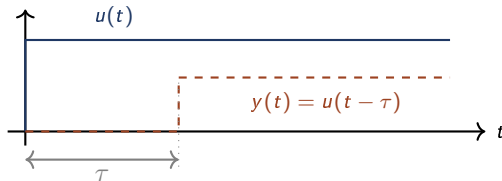
Same idea, different picture — the map $z = e^{sT}$ sends the imaginary axis to the unit circle.

FEE3411 works entirely in continuous time; this comparison is here because the syllabus asks for it — and because every controller you actually build is a discrete one running on a processor.

Pure time delay

A conveyor belt, a pipe, a communication link, a computation — the output is the input, unchanged in shape, τ seconds later.

$$y(t) = u(t - \tau)1(t - \tau) \quad \Rightarrow \quad \frac{Y(s)}{U(s)} = e^{-\tau s}.$$



Delay breaks the polynomial methods

Common pitfall

$e^{-\tau s}$ is *not* rational, so a system with a pure delay has no finite set of poles and zeros and cannot be put on a pole-zero map. Every method resting on a polynomial characteristic equation — Routh–Hurwitz (Week 7), root locus (Week 8) — does not apply directly. The frequency-domain methods of Weeks 10–12 do: $e^{-j\omega\tau}$ has magnitude 1 and phase $-\omega\tau$, simply added to the phase plot. Hence Nise treats delay in §10.12 (p. 597), inside the Bode chapter.

Where a rational approximation is needed, the first-order **Padé approximation** (Khalil, p. 14):

$$e^{-\tau s} \approx \frac{1 - \tau s/2}{1 + \tau s/2},$$

accurate to the s^2 term. Its zero sits at $s = +2/\tau$ — *in the right half plane*: a delay behaves like a non-minimum-phase system, which is why it is so damaging to a loop.

Delay time is a different thing

Common pitfall

The syllabus phrase *delay time* does not mean the pure time delay τ . Delay time t_d is a **transient specification** — the time a step response takes to reach 50% of its final value — defined in Week 5 for systems with no transport lag at all.

A system can have a delay time of 0.4 s and *no* time delay whatsoever. Read from the context which one a question means.

Pure time delay τ	Delay time t_d
Transport lag: output is input, later	Transient spec: time to reach 50%
Transfer function $e^{-\tau s}$, not rational	Read off a step response curve
This week	Week 5

Summary and what comes next

Summary — getting a transfer function

- The Laplace transform turns a linear constant-coefficient ODE into algebra, carrying initial conditions in through $\mathcal{L}\{\dot{f}\} = sF(s) - f(0^-)$.
- $G(s) = Y(s)/U(s)$ with zero initial conditions is a property of the plant alone. Its denominator is the characteristic polynomial, its inverse transform is the impulse response, and cascaded blocks multiply.
- For a network, the impedances R , Ls , $1/Cs$ give the transfer function directly by divider algebra — no differential equation needed.

Summary — reading a transfer function

- Poles are the denominator roots, zeros the numerator roots; the pole-zero map is symmetric about the real axis.
- Each pole contributes one mode: real part sets growth or decay, imaginary part sets oscillation. Stable $\iff$ every pole in the open left half plane.
- Zeros add no modes — they reweight the residues. A right-half-plane zero causes initial undershoot but never instability.
- D.c. gain is $G(0) = b_0/a_0$, not the gain factor K ; the final value theorem gives it, but only for a stable system.
- Discrete systems use $G(z)$ with the unit circle as the stability region. A pure delay is $e^{-\tau s}$, is not rational, and is not the same thing as delay time.

Before the tutorial

You should be able to attempt these before you walk in (notes, “Before the tutorial”):

- 1 Using impedances only, find $V_o(s)/V_i(s)$ for a series RC network with the output across the resistor. Give its pole, zero and d.c. gain, and say what kind of filter it is.
- 2 A system has poles at $-2, -5$, a zero at -10 and d.c. gain 3. Write $G(s)$ in factored and polynomial form, and state its relative degree.
- 3 Find the step response of $G(s) = 6/[(s+1)(s+3)]$ by partial fractions and check it at $t = 0$ and $t \rightarrow \infty$.
- 4 Without computing: sketch the natural response for poles at $-0.5 \pm j8$, and say which feature the -0.5 sets and which the 8 sets.

The tutorial itself works full transfer-function derivations for RLC and mass–spring–damper systems, more inverse Laplace by partial fractions, and the link from poles and zeros to response shape.

Looking ahead

- **Week 4** — how transfer functions combine when subsystems are wired together: block diagrams, canonical feedback form, reduction rules, signal flow graphs and Mason's gain formula. It leans hardest on this week's impedance shortcut and cascade rule.
- **Assignment 2** (issued Week 4, due Week 6) covers Weeks 3 and 4 together — this week's partial-fraction and transfer-function work is examined alongside next week's material, so consolidate it now.
- **Beyond** — the pole-zero map is the picture the rest of the unit is drawn on: Week 5 reads transient specs off pole positions, Week 7 tests whether a pole has crossed the imaginary axis, Week 8 tracks how poles move with gain.

The boundary, stated plainly

Locating and interpreting poles is **analysis** — FEE3411. Choosing a compensator to put them somewhere better is **design** — FEE3412.

Reading for this week

Primary

Khalil, §2-1 Differential Equation and Transfer Function Models (pp. 10–14); App. A, Review of the Laplace Transform (p. 438); App. C, Laplace and z-Transform Tables (p. 444).

Secondary

Nise (6th ed.), §2.3 The Transfer Function (p. 44); §4.2 Poles, Zeros, and System Response (p. 162); §10.12 Systems with Time Delay (p. 597, concept only).

Further

Sundararajan, §2.1–2.3.